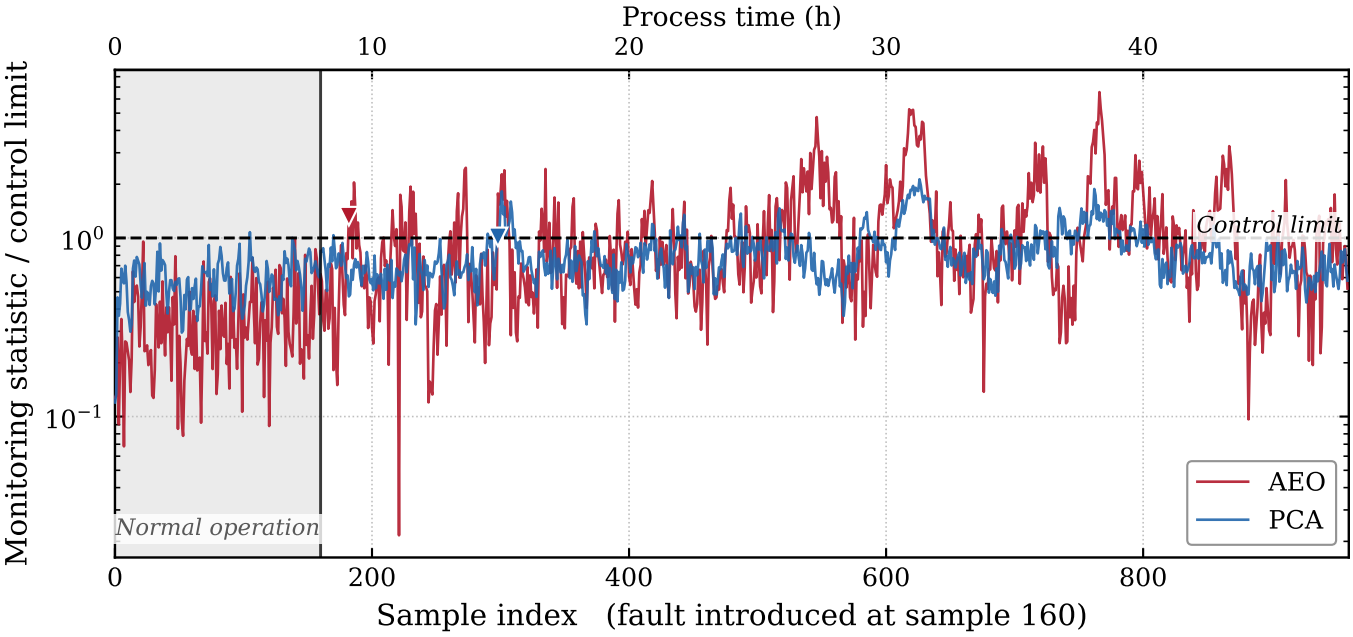
Fault Detection Performance for IDV(10)

Hotelling's T^2 statistic — AEO vs. PCA at a common 1% false-alarm rate



Held-out performance (mean of 7 independent runs)

AEO	delay	84.7 samples (254 min)	MDR 0.653	detected in 7/7 runs
PCA	delay	201.6 samples (605 min)	MDR 0.777	detected in 7/7 runs

AEO detects 116.9 samples (351 min) earlier; AEO missed-detection rate lower by 16%